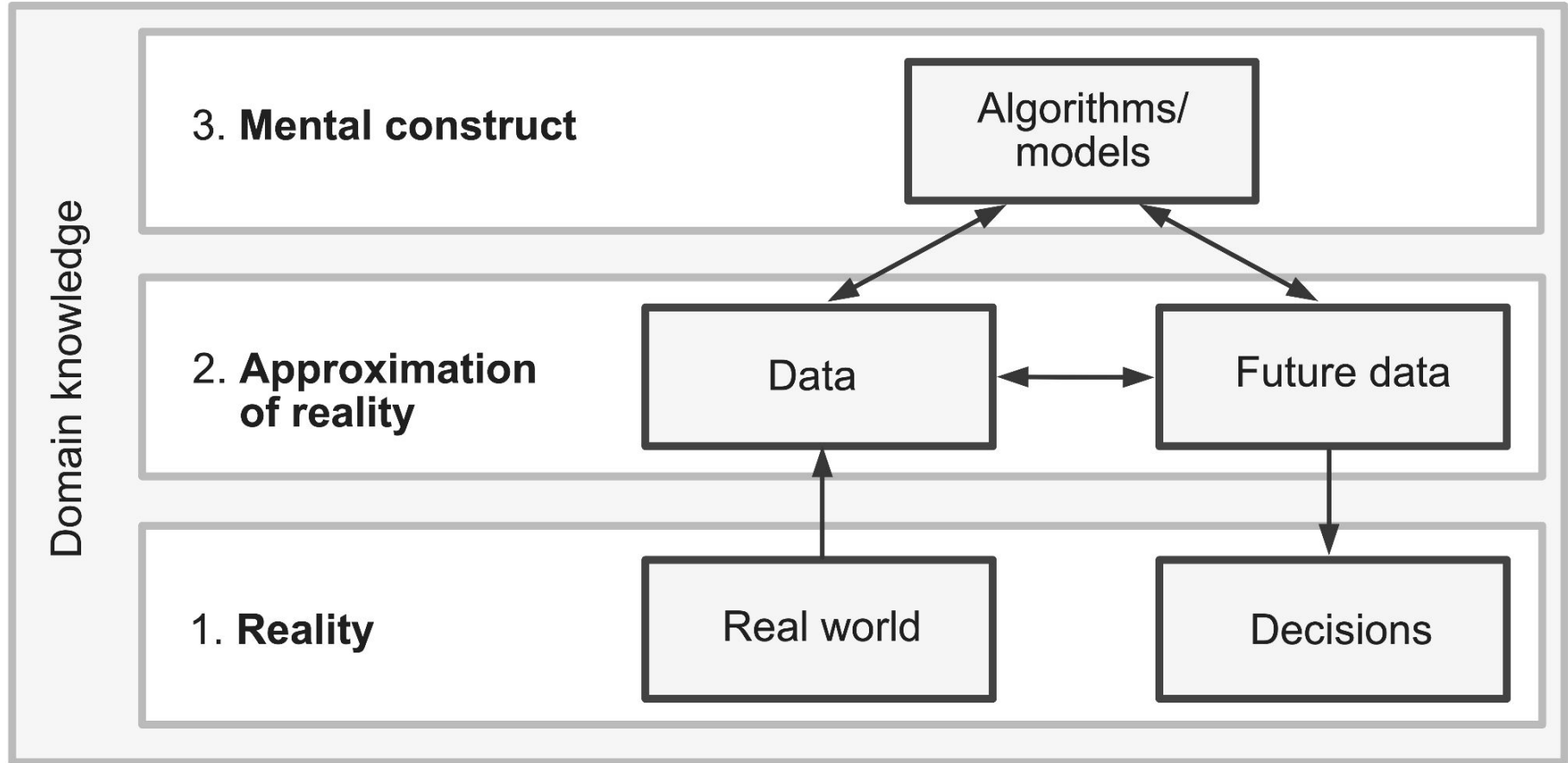


DS-3001

Data Analysis and Visualization

Data Analysis Lifecycle

The Role of Data and Algorithms in Real-World Decision Making



Role of Critical Thinking

- Asking questions about the domain problem.
 - Is the question being framed really the question you care about,
 - and is this question ethical in the first place?

e.g.

 - “Which features of our website can we change that will make our customers less likely to cancel their subscription?”

VS

 - “Why do customers tend to cancel our product subscription?”

Role of Critical Thinking

- Asking questions about the data.
 - Where did the data come from?
 - Is the source of the data trustworthy?
 - Is the data relevant to the question being asked, as well as any future setting to which the results might be applied?
 - What do the values contained within the data actually correspond to in the real world?

The ROCCC framework

Principle	Core Question	What It Evaluates
Reliable	Is the data accurate and error-free?	Evaluates whether the data collection process was rigorous, unbiased, and vetted by a trusted organization.
Original	Can you locate the primary source?	Checks whether the dataset comes directly from the entity that gathered it, rather than a second-hand aggregator where errors or context loss can occur.
Comprehensive	Does it contain all necessary details?	Assesses whether the dataset provides all critical parameters required to answer the business question without major missing values or sampling bias.
Current	Is the data up-to-date?	Determines if the information reflects present conditions, as outdated data loses utility and can lead to obsolete conclusions.
Cited	Is the source properly documented?	Verifies that metadata exists regarding who collected the data, when it was created, and how variables are defined.

Role of Critical Thinking

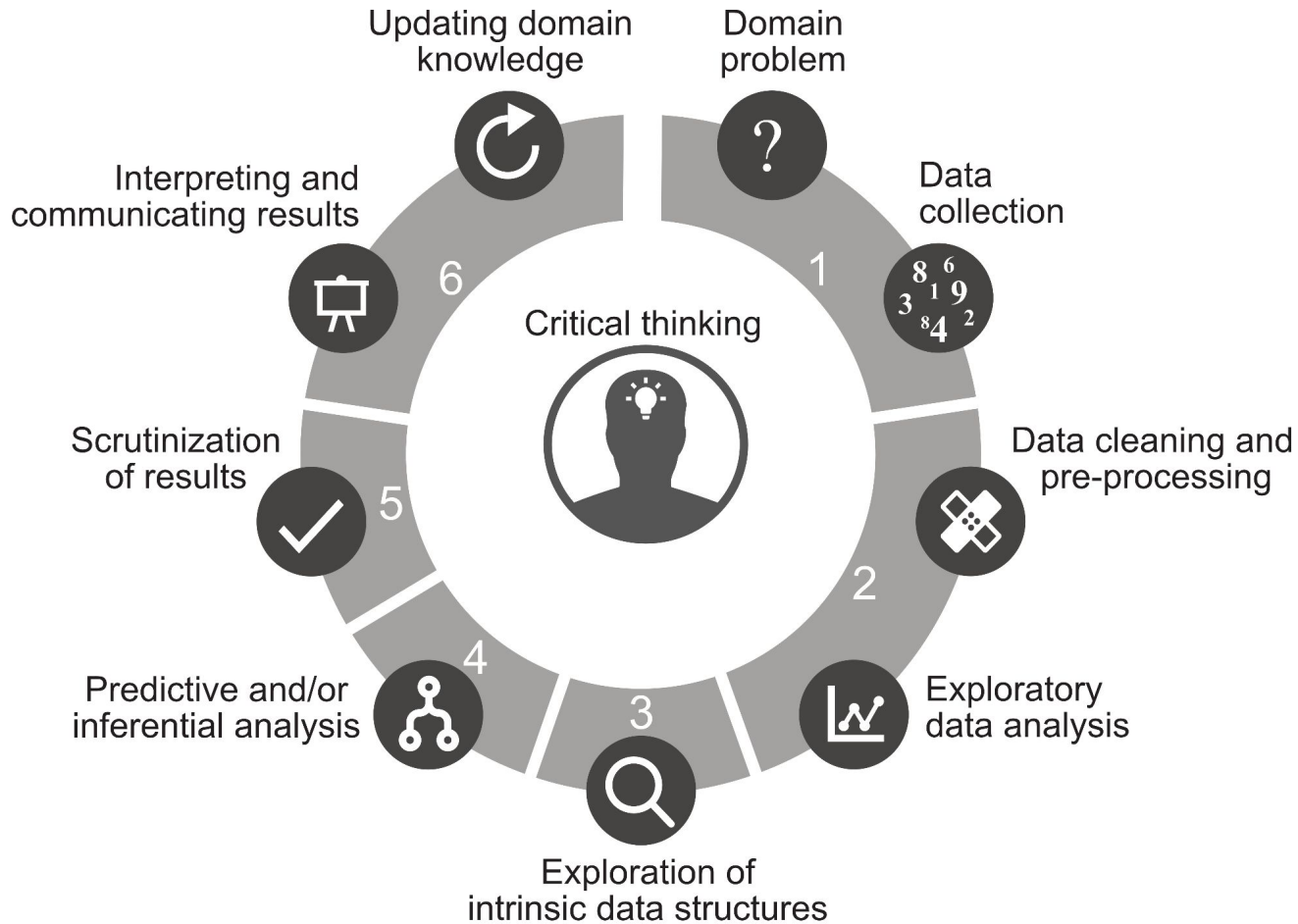
- Asking questions about the analysis, algorithms, and results.
 - What do these results mean in the context of reality?
 - Do they make sense?
 - Do the analysis and algorithms involve making any assumptions about the data or domain?
 - Are these assumptions reasonable?
 - Do these results answer the primary question?
 - Are the results being communicated accurately?

Role of Critical Thinking

- Confirmation Bias
 - when you fail to critically scrutinize results because they match what you expect to find
- Desirability Bias
 - when you try to find the results that you think other people (or popular opinion) expect you to find

Evaluating and Building Trustworthiness Using the PCS Framework

- **Predictability.**
 - demonstrating that the results hold in the context of the actual future or external data to which you will be applying your results.
 - Training, Validation and Test Sets
- **Computability.**
 - computations are clear, efficient, scalable, and reproducible
- **Stability.**
 - Your conclusions are considered stable if you can show that they remain fairly consistent when realistic perturbations (changes/modifications) are made to the
 - underlying data, judgment calls, and, algorithmic choices.



Formulating the Right Question

- **Collaborate with Domain Experts:** Translate broad domain goals into precise, answerable data questions.
- **Identify the Core Objective:** Clarify the "why" behind the question (e.g., predicting patient risk versus uncovering systemic hospital flaws).
- **Align with Available Data:** Continually refine and adapt the project scope based on the reality of what data is actually accessible.

Data Collection Protocol

- **Assess Data Sources:** Determine if relying on existing public/internal data is sufficient, or if new, manual collection is required.
- **Involve Analysts Early:** Data scientists must help design collection protocols; data collected without their input is often poorly suited for analysis.
- **Understand Data Structure:** Anticipate tabular (rectangular) layouts, distinguishing between observational units (rows) and variables/features (columns).

Making a Plan for Evaluating Predictability

- If you can **collect additional data** (which resembles the future data to which you will be applying your results or algorithms), then you should either collect it now or make a plan for collecting it later.
- If not, you may need to **split your existing data** into a training set and validation set (and a test set, if relevant)

Data Cleaning & Preprocessing

- **Format Conversion:** Transform raw text, images, or unstructured formats into structured, rectangular datasets.
- **Standardize Variable Types:** Categorize data into numeric, categorical, date/time, and structured/unstructured text features.
- **Iterative Process:** Revisit this stage frequently as hidden errors, missing values, or data anomalies are discovered later in the cycle.

Uncovering Intrinsic Data Structures

- **Optional but Powerful:** Not every project needs this stage; its use depends entirely on the complexity of the data and the initial question.
- **Dimensionality Reduction:** Data be projected onto a lower-dimensional subspace that summarizes the most informative parts of the original data.
- **Clustering:** Discover natural, hidden groupings within observational units to identify new patterns without relying on pre-labeled categories.

Building Models & Drawing Inferences

- **Predictive Modeling:** Train algorithms (such as Least Squares, Logistic Regression, Random Forests, or Decision Trees) to forecast continuous or binary outcomes.
- **Data-Driven Inference:** Draw robust statistical conclusions about underlying population mechanisms or relationships.
- **Question-Driven Applicability:** Like Stage 3, modeling is strictly dictated by whether the initial domain question requires prediction or inference.

Statistical Inference Alignment

- **Algorithmic Realm:** Assumes probabilistic data-generating processes (e.g., normal distribution, independent and identically distributed (i.i.d.) sampling, and well-defined hypothesis boundaries.
- **Data Realm Alignment:** Assumes the observed dataset reflects an unbiased random sample free from latent confounding, continuous measurement errors, or systematic missingness.
- **Real-World Realm Alignment:** Assumes the underlying system operates under stable physical, biological, or social mechanisms that remain invariant to measurement methods.

Learning Models - Linear / Logistic Regression

- **Algorithmic Realm:** Assumes direct linearity, independence of errors.
- **Data Realm Alignment:** Highly sensitive to outliers, and measurement noise in tabular datasets.
- **Real-World Realm Alignment:** Works best when physical or economic laws approximately exhibit proportional relationships (e.g., dosage vs. baseline physiological response, supply vs. price elasticity).
- **Pitfall:** If the real world possesses non-linear threshold effects or complex interactions, a linear model enforces a mathematical construct that fails to capture the underlying reality.

Tree-Based Ensembles (Random Forests, XGBoost)

- **Algorithmic Realm:** Non-parametric recursive partitioning; minimizes loss by grouping regions of feature space.
- **Data Realm Alignment:** Handles mixed data types, missing values, and high dimensionality effectively without requiring formal transformations.
- **Real-World Realm Alignment:** Suitable for complex domain phenomena with threshold behaviors (e.g., medical triage guidelines, risk scoring).
- **Pitfall:** Tree ensembles are step-function estimators and struggle to extrapolate continuous trend lines into unobserved feature space outside the training distribution.

Deep Neural Networks (CNNs, Transformers, LLMs)

- **Algorithmic Realm:** High-capacity non-linear function approximators, overparameterized manifolds, self-attention.
- **Data Realm Alignment:** Excels at extracting structure from high-dimensional, unstructured data (pixels, raw audio, token streams).
- **Real-World Realm Alignment:** Effective when physical relationships are hierarchical or local (e.g., visual feature composition in computer vision or spatial dependencies in climate modeling).
- **Pitfall:** Deep networks can easily fit non-causal spurious correlations present in the Data Realm (e.g., classifying a hospital's watermark instead of a pathology feature on an X-ray).

Evaluation of Results

- Qualitatively
 - using critical thinking
- Quantitatively
 - using PCS framework
- Present to Domain experts
 - Use Negative Control

Communication of Results & Updating Domain Knowledge

- **Bridge the Gap:** Translate highly technical and mathematical findings back into accessible language for domain experts and stakeholders.
- **Update the Field:** Use the veridical (truthful/accurate) insights gained to update collective domain knowledge and influence practical decision-making.
- **Acknowledge the Non-Linear Path:** Document and present the findings not as a straight line, but as the result of a dynamic, iterative life cycle to support future reproducibility.

References

1. Veridical Data Science by Bin Yu and Rebecca L. Barter - Chapter 1 & 2
 - <https://vdsbook.com/>